

Con. 7749-13.**LJ-13927****(3 Hours)****[Total Marks : 100****N.B. :** (1) Question No. 1 is **compulsory**.(2) Attempt any **four** questions out of remaining **six** questions.(3) **Figures** to the **right** indicate **full** marks.

1. (a) Model the following as a fuzzy set using suitable membership function - "Numbers close to 6".
(b) Explain standard fuzzy membership functions.
(c) Determine all α - level sets and strong α -level sets for the following fuzzy set.
 $A = \{ (1, 0.2) (2, 0.5), (3, 0.8), (4, 1), (5, 0.7), (6.03) \}$.
2. Design a Fuzzy Controller to determine the the wash time of a domestic washing machine. Assume that the inputs are dirt and grease on the clothes. Use three descriptors for each input variable and five descriptors for output variable. Derive a set of rules for control action and defuzzification. The design should be supported by figures wherever possible. Clearly indicate that if the clothes are soiled to a larger degree the wash time required will be more.
3. (a) Determine the weights after four steps of training for Perceptron learning rule of a single neuron network starting with initial weights :-
 $W = [0 \ 0]^t$, inputs as $X_1 = [2 \ 2]^t$,
 $X_2 = [1 \ -2]^t$, $X_3 = [-2 \ 2]^t$, $X_4 = [-1 \ 1]^t$,
 $d_1 = 0$, $d_2 = 1$, $d_3 = 0$, $d_4 = 1$ and $c = 1$.
(b) Explain Mamdani type of Fuzzy Inference system in detail.
4. (a) Prove the following identities :-
(i) For unipolar continuous activation function
 $f^1(\text{net}) = 0 \ (1 - 0)$
(ii) For bipolar continuous activation function :-
$$f^1(\text{net}) = \frac{1}{2}(1 - 0^2)$$

(b) Explain error back propagation training algorithm with the help of a flowchart.
5. (a) Explain RBF network and give the comparison between RBF and MLP.
(b) Explain with examples linearly and non-linearly separable pattern classification.
6. (a) What is learning in neural networks ? Differentiate between Supervised and Unsupervised Learning.
(b) Explain Travelling salesperson problem using simulated annealing.
7. Write notes on any **two** of the following :-
(a) Learning Vector Quantization.
(b) Derivative Free Optimization.
(c) Winner take all learning rule.